UNIVERSITY ELEPHANT MADA
FACULTY OF MATHEMATICS AND NATURAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ELECTRONICS
IUP COMPUTER SCIENCE

FINAL EXAMINATION
Semester II Academic Year 2024/2025

Course name : Machine Learning - CSA
Day, date : Wednesday , 11 June 2025
Time : 07.30 – 09.20 (110 minutes)
Lecturer : Afiahayati, M.Cs. , Ph.D.
Exam Type : **Open 1 sheet of A4 hand written notes, Open Calculator ,** Closed Book , Gadget OFF, Laptop OFF, Internet OFF

Instructions:

1. **This final exam contributes 35% of your final grade**
2. **Please submit your A4 handwritten notes together with the answer sheet**

PLO2; CO-2: Students are able to understand data preparation and performance evaluation in machine learning (score: 10%).

1. Please answer the questions below !
   a. Look at Figure 1. Please write the output of the python code below , if the total number of data from the dataset is 500! (*score : 2%)*

```
[ ] from sklearn.model_selection import train_test_split

    X_train, X_test, y_train, y_test = train_test_split(X, y, train_size = 0.7)

[ ] print("Banyak data latih setelah dilakukan Train-Test Split: ", len(X_train))
    print("Banyak data uji setelah dilakukan Train-Test Split: ", len(X_test))
```

Figure 1. Code Snippets of Data preparation

   b. Please explain about k- *fold cross validation* ! When should we use k- *fold cross validation* ? (*score : 3%)*
   c. Look at Table 1. Table 1 represents the target and prediction results of a classification model. There are 2 classes , namely the YES class and the NO class . Create a Confusion Matrix from the classification results , then calculate the accuracy, sensitivity / recall , precision and F1-measure! (score : 5%)

Table 1. Classification results table

No.	Target	Classification Results	
1	NO	NO	TN = 3
2	NO	YES	FP = 1
3	YES	YES	TP = 4
4	YES	NO	FN = 4
5	YES	YES	TP
6	NO	NO	TN
7	NO	NO	TN
8	YES	NO	FN
9	YES	YES	TP
10	YES	NO	FN
11	YES	NO	FN
12	YES	YES	TP

PLO3; CO-5: Students are able to understand k-NN and apply it to solve a case (score : 4%).

2. Determine whether the following statements are correct or wrong and give the reason
   a. K-NN is more suitable used on large datasets because its high speed for prediction. (score :2%)
   b. K-NN must use the *Euclidian Distance* formula (score 2%)

PLO3; CO-6: Students are able to understand Naive Bayes Model and apply it to solve a case (score : 5%).

3. Given data as following (Table 2)

Table 2. Play Tennis Data

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

w-8
mild = 6
Rain = 5
normal = 7
yes = 9
No = 5

a. Create a probability table from the playtennis data above. Then use the Naive Bayes Model to determine whether on days D15 and D16 it is advisable to play tennis or not with the following weather. (Score : 3,5%)
   D15: Outlook = Rain, Temperature = Mild, Humidity = Normal, Wind = Weak

b. In the beginning, Naïve bayes is designed for the categorical data. But Naïve bayes model also can be implemented for numerical data. Please explain, how is Naïve bayes implemented for numerical data? (Score : 1,5%)

PLO3; CO-7: Student are able to understand ensemble learning and apply it to solve a case (score 10%)

4. a. Please explain the algorithms of bagging, boosting and stacking! Please also explain the diffencess! (score : 3%)

   b. Look at Figure 2. Please explain what is the meaning of base classifier, n_estimators and learning rate! (score:2%)

```
adaboost_classifier = AdaBoostClassifier(
    base_classifier, n_estimators=50, learning_rate=1.0, random_state=42
)
adaboost_classifier.fit(X_train, y_train)
```

Figure 2. Code Snippets of Adaboost

3.888
0.6
1.6530

✓c. Please pay attention to the following data. Please implement the Bagging algorithm for the data. Please use only 3 bootstrapings/single models! Please write the final prediction and calculate the accuracy! (score : 5%)

Table 3. Data for Bagging Algorithm

Source Dataset								
	x	2,4	3,1	3,7	4,4	5,2	5,8	6,4
	y	1	1	1	-1	-1	-1	-1

bagging 1
I 2
bagging 3

PLO3; CO-7: Student are able to understand clustering and apply it to solve a case (score 6%)

5. ✓a. Please explain what is the difference between clustering and classification? (score: 3%)

b. Please explain how to determine the optimal value of k in k-means clustering. Please loot at Figure 3. Based on the figure 3, please determine the optimal value of k! (score : 3%)

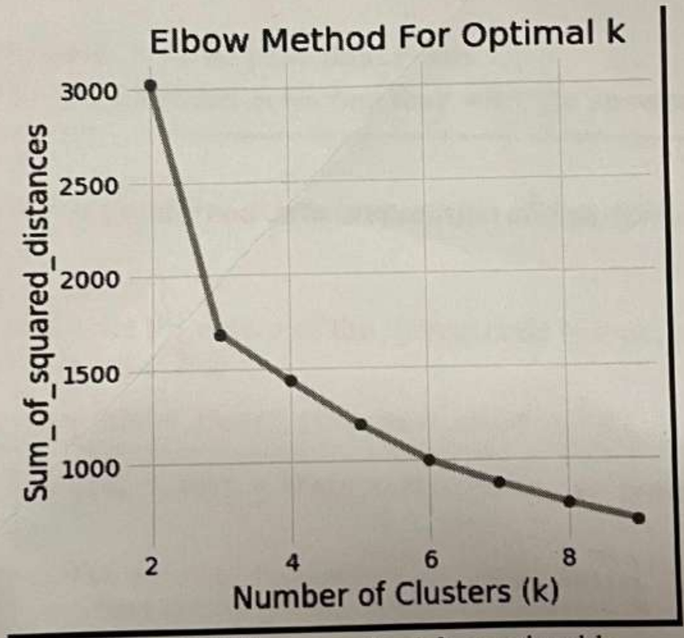

Figure 3. Elbow method for optimal k

-- No effort will be vain –